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Security Quiz 2

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Result of Security Quiz 2:

Total Questions	Full Score	Passing Rate	Your Score	Correct Answer Percentage	Elapsed
8	90	80%	90	100%	00:00:52

Congratulations!

You passed this test!

If you want to retake this quiz, please press Ctrl + F5 on Windows or press CMD + R on Mac.

Your answers are shown below:

Question 1

What are two recommendations for protecting network ports from being exploited when located in an office space outside of an IT closet? (Choose two)

- A. configure ports to a fixed speed
- B. enable the PortFast feature on ports
- C. configure static ARP entries
- **D. implement port-based authentication**
- **E. shut down unused ports**

Question 2

Which technology is used to improve web traffic performance by proxy caching?

- A. ASA
- B. Firepower
- C. FireSIGHT
- **D. WSA**

Explanation

The Web Security appliance (WSA) intercepts requests that are forwarded to it by clients or other devices over the network. Proxy caching is a setting in WSA that caches data to increase performance.

Question 3

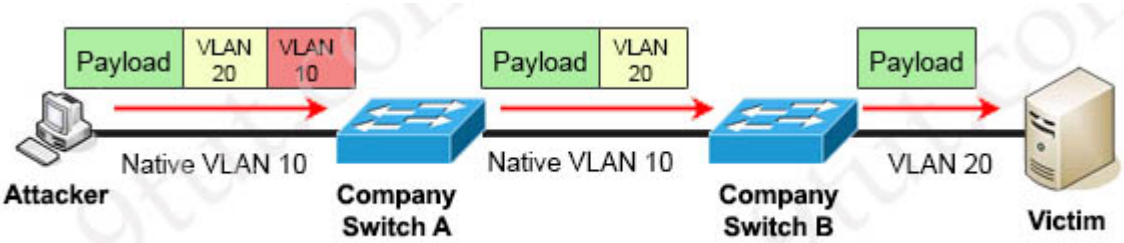
What is a practice that protects a network from VLAN hopping attacks?

- A. Enable dynamic ARP inspection
- B. Configure an ACL to prevent traffic from changing VLANs
- C. Implement port security on internet-facing VLANs
- **D. Change native VLAN to an unused VLAN ID**

Explanation

VLAN Hopping: By altering the VLAN ID on packets encapsulated for trunking, an attacking device can send or receive packets on various VLANs, bypassing Layer 3 security measures. VLAN hopping can be accomplished by switch spoofing or double tagging. One of a popular type of VLAN Hopping is

Double-Tagging attack:



In this attack, the attacking computer generates frames with two 802.1Q tags. The first tag matches the native VLAN of the trunk port (VLAN 10 in this case), and the second matches the VLAN of a host it wants to attack (VLAN 20).

When the packet from the attacker reaches Switch A, Switch A only sees the first VLAN 10 and it matches with its native VLAN 10 so this VLAN tag is removed. Switch A forwards the frame out all links with the same native VLAN 10. Switch B receives the frame with an tag of VLAN 20 so it removes this tag and forwards out to the Victim computer.

Note: This attack only works if the trunk (between two switches) has the same native VLAN as the attacker. In other words, this attack is only successful if the attacker belongs to the native VLAN of the trunk link. Another important point is, this attack is strictly one way as it is impossible to encapsulate the return packet.

To mitigate this type of attack, we can use VLAN access control lists (VACLs, which applies to all traffic within a VLAN. We can use VACL to drop attacker traffic to specific victims/servers); or implement Private VLANs; or keep the native VLAN of all trunk ports different from user VLANs.

Question 4

Refer to the exhibit. An engineer booted a new switch and applied this configuration via the console port. Which additional configuration must be applied to allow administrators to authenticate directly to enable privilege mode via Telnet using local username and password?

```
Switch(config)#hostname R1
R1(config)#interface FastEthernet0/1
R1(config-if)#no switchport
R1(config-if)#ip address 10.100.20.42 255.255.255.0
```

```
R1(config-if)#line vty 0 4
R1(config-line)#login
```

- **A. 1(config)#username admin privilege 15 secret p@ss1234**
  R1(config-if)#line vty 0 4
  R1(config-line)#login local
- B. R1(config)#username admin secret p@ss1234
  R1(config-if)#line vty 0 4
  R1(config-line)#login local
  R1(config)#enable secret p@ss1234
- C. R1(config)#username admin
  R1(config-if)#line vty 0 4
  R1(config-line)#password p@ss1234
- D. R1(config)#username admin
  R1(config-if)#line vty 0 4
  R1(config-line)#password p@ss1234
  R1(config-line)#transport input telnet

Explanation

With "login local" command, the router will use the local username and password for authentication. And we assigned the highest privilege level of 15 for this user so we are put into privilege mode automatically (R1#).

Question 5

An organization secures its network with multi-factor authentication using an authenticator app on employee smartphones. How is the application secured in the case of a user's smartphone being lost or stolen?

- **A. The application requires the user to enter a PIN before it provides the second factor.**
- B. The application challenges a user by requiring an administrator password to reactivate when the smartphone is rebooted.
- C. The application requires an administrator password to reactivated after a configured interval.
- D. The application verifies that the user is in a specific location before it provides the second factor.

Question 6

Which type of attack can be mitigated by dynamic ARP inspection?

- A. malware
- B. DDoS
- **C. man-in-the-middle**
- D. worm

Explanation

ARP attack (like ARP poisoning/spoofing, man-in-the-middle) is a type of attack in which a malicious actor sends falsified ARP messages over a local area network as ARP allows a gratuitous reply from a host even if an ARP request was not received. This results in the linking of an attacker's MAC address with the IP address of a legitimate computer or server on the network. This is an attack based on ARP which is at Layer 2. Dynamic ARP inspection (DAI) is a security feature that validates ARP packets in a network which can be used to mitigate this type of attack.

Question 7

Which device performs stateful inspection of traffic?

- A. wireless controller
- B. switch
- **C. firewall**
- D. access point

Explanation

In stateful inspection, the firewall not only inspects packets up through the application layer/layer7 determining a packet's header information and data content, but also monitors and keeps track of the connection's state. For all active connections traversing the firewall, the state information, which may include IP addresses and ports involved, the sequence numbers and acknowledgement numbers of the packets traversing the connection, TCP packet flags, etc. is maintained in a state table.

Question 8

Which technology can prevent client devices from arbitrarily connecting to the network without state remediation?

- A. 802.11n
- B. MAC Authentication Bypass
- C. IP Source Guard
- **D. 802.1x**

Explanation

The IEEE 802.1x standard defines a client-server-based access control and authentication protocol that prevents unauthorized clients from connecting to a LAN through publicly accessible ports unless they are properly authenticated. The authentication server authenticates each client connected to a switch port before making available any services offered by the switch or the LAN. Until the client is authenticated, 802.1x access control allows only Extensible Authentication Protocol over LAN (EAPOL), Cisco Discovery Protocol (CDP), and Spanning Tree Protocol (STP) traffic through the port to which the client is connected. After authentication is successful, normal traffic can pass through the port.

Comments (3) Comments

1. CCNP in process
   June 20th, 2024

   Telnet question is wrong, I mean you need to enable telnet with cmd transport input right? The "right one" doesn't enable it just set auth to vty lines

2. @CCNP in process
   June 26th, 2024

   Imo yes, but all other answers are "more wrong"

3. @CCNP in process
   July 3rd, 2024

Not necessarily, depending on the device telnet might be enabled by default. At least from what I have seen.

Regardless, Telnet is heavily discouraged.

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VPN Quiz Port Security Quiz

Premium Member Zone

Welcome Darène MAYISSA!

- Welcome Premium Member!
- CCNA - New Questions Part 5
- CCNA - New Questions Part 6
- CCNA - New Questions Part 7
- CCNA - New Questions Part 8
- CCNA - New Questions Part 9
- CCNA - New Questions Part 10
- CCNA - New Questions Part 11
- CCNA - New Questions Part 12
- CCNA - New Questions Part 13
- CCNA - New Questions Part 14
- CCNA - New Questions Part 15
- CCNA - New Questions Part 16
- Composite Quizzes
- === New CCNA v1.1 Lab Sims (very important) ===
- Voice VLAN Configuration Sim 2
- Voice VLAN & LLDP Sim
- Allowed, Native VLAN & LACP Sim
- Static Routing Configuration Sim 4
- IPv4 and IPv6 Connectivity Sim
- VLAN & CDP Sim 2
- IPv4 and IPv6 Assignment Sim
- VLAN CDP & LLDP Sim
- OSPF Configuration Sim 2
- OSPF Configuration Sim 3
- Trunking & LACP Sim
- VLAN Configuration Sim
- Allowed, Native VLAN & LACP Sim 2
- 802.1Q Trunking & LACP
- IPv4 and IPv6 Assignment Sim 2
- 802.1Q Trunking, Native VLAN & LACP
- Static Routing Configuration Sim 5
- Static Routing Configuration Sim 6
- PAT, NTP & DHCP Relay Sim

- OSPF DR Sim
- NACL DHCP Snooping 3
- NAT DHCP SSH Sim
- ==== Lab Sims for CCNA (although they are rather old but please learn all of them too) ====
- IP Services Sim Version 2
- OSPF Configuration Sim
- LACP Configuration Sim
- Voice VLAN Configuration Sim
- VLAN and Trunking Configuration Sim
- Named Access-list & Port Security Sim
- Named Access-list & DHCP Snooping Sim
- VLAN and CDP Sim
- IPv4 and IPv6 Deployment Sim
- Static Routing Configuration Sim 3
- VLAN & LLDP Sim
- IPv6 Static Routing Sim
- Named Access-list & DHCP Snooping Sim 2
- ISL Trunking and LACP Sim
- === Below Sims are old and retired so you do not need to learn them ===
- Static Routing Configuration Sim 2
- IP Services Sim
- Static Routing Configuration Sim

CCNA Training

- CCNA Knowledge
- CCNA Tutorials, Practice Labs & Lab Challenges
- Spine-Leaf Architecture Tutorial
- Generative AI vs Predictive AI Tutorial
- REST API Tutorial
- Network Time Protocol (NTP) Tutorial
- DHCP Snooping Tutorial
- What is SDN?
- How a router makes a forwarding decision
- Configure SSH for Remote Access on Cisco Router
- Key Security Concepts Tutorial
- Forwarding per-hop behavior (PHB) for QoS Tutorial
- Port Security Tutorial
- Puppet Tutorial
- Chef Tutorial
- Ansible Tutorial
- JSON Tutorial
- Layer 2 Threats and Security Features
- AAA TACACS+ and RADIUS Tutorial
- STP Root Port Election Tutorial
- TCP and UDP Tutorial
- WAN Tutorial
- DHCP Tutorial
- Simple Network Management Protocol SNMP Tutorial
- Syslog Tutorial
- EtherChannel Tutorial
- Hot Standby Router Protocol HSRP Tutorial
- InterVLAN Routing Tutorial
- Cisco Command Line Interface CLI
- Cisco Router Boot Sequence Tutorial
- OSI Model Tutorial
- Subnetting Tutorial – Subnetting Made Easy

- Frame Relay Tutorial
- Wireless Tutorial
- Virtual Local Area Network VLAN Tutorial
- VLAN Trunking Protocol VTP Tutorial
- IPv6 Tutorial
- Rapid Spanning Tree Protocol RSTP Tutorial
- Spanning Tree Protocol STP Tutorial
- Network Address Translation NAT Tutorial
- Access List Tutorial
- RIP Tutorial
- EIGRP Tutorial
- OSPF Tutorial
- Practice CCNA GNS3 Labs
- CCNA Lab Challenges

CCNA 200-301

- CCNA FAQs & Tips
- Basic Questions
- Topology Architecture Questions
- Cloud & Virtualization Questions
- CDP & LLDP Questions
- Switch Questions
- VLAN & Trunking Questions
- VLAN & Trunking Questions 2
- STP & VTP Questions
- EtherChannel Questions
- TCP & UDP Questions
- IP Address & Subnetting Questions
- IP Routing Questions
- IP Routing Questions 2
- OSPF Questions
- OSPF Questions 2
- EIGRP Questions
- NAT Questions
- NTP Questions
- Syslog Questions
- HSRP Questions
- Access-list Questions
- AAA Questions
- Security Questions
- Security Questions 2
- DAI Questions
- IPv6 Questions
- DNS Questions
- QoS Questions
- Port Security Questions
- Wireless Questions
- Wireless Questions 2
- SDN Questions
- DNA Center Questions
- Drag Drop Questions
- Drag Drop Questions 2
- Drag Drop Questions 3
- VPN Questions

- DHCP Questions
- Automation Questions
- Miscellaneous Questions
- Share your CCNA Experience

Network Resources

- Free Router Simulators
- CCNA Website
- ENCOR Website
- ENSDWI Website
- ENARSI Website
- DevNet Website
- CCIE R&S Website
- Security Website
- Wireless Website
- Design Website
- Data Center Website
- Service Provider Website
- Collaboration Website

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